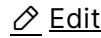


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Absorbing exceptional points in the time domain

This set of notes is on the paper: [Excitation of absorbing exceptional points in the time domain](https://journals.aps.org/prl/pdf/10.1103/PhysRevA.106.L031503) (<https://journals.aps.org/prl/pdf/10.1103/PhysRevA.106.L031503>) by **Farhi et al.**

Section 1: Derivation of Equations 1 and unnumbered equation between Equation 1 and 2

In this section, we derive **Equation 1** and the unnumbered equation between **Equation 1** and **Equation 2** in the main text of the paper. Note that we use the standard physicist definitions for the Fourier transforms, so some factors may be off from the paper.



$$\begin{aligned}
F[f](\omega) &= \int_{-\infty}^{\infty} f(t)e^{i\omega t} dt \rightarrow F[e^{-i\omega_1 t}] = 2\pi\delta(\omega - \omega_1) \\
F[te^{-i\omega_1 t}] &= \int_{-\infty}^{\infty} te^{-i\omega_1 t} e^{i\omega t} dt = -i\partial_{\omega} \left[\int_{-\infty}^{\infty} e^{i(\omega - \omega_1)t} dt \right] = -2\pi i\partial_{\omega}\delta(\omega - \omega_1) \\
\frac{1}{2\pi} \int_{-\infty}^{\infty} \rho(\omega) \left[-2\pi i\partial_{\omega}\delta(\omega - \omega_1) \right] e^{-i\omega t} d\omega &= \\
-i \int_{-\infty}^{\infty} \left[\partial_{\omega} \left[\rho(\omega)\delta(\omega - \omega_1)e^{-i\omega t} \right] + it\delta(\omega - \omega_1)\rho(\omega)e^{-i\omega t} - \delta(\omega - \omega_1)e^{-i\omega t}\partial_{\omega}\rho(\omega) \right] d\omega &= \\
= t\rho(\omega_1)e^{-i\omega_1 t} + ie^{-i\omega_1 t}\partial_{\omega}\rho(\omega_1), &
\end{aligned}$$

where in the last step we used the fact that $-\infty < \omega_1 < \infty$ which implies that $\delta(\omega \pm \infty) = 0$.

Effect of turn-on of the input wave (Unnumbered equation between Equation 1 and Equation 2)

$$\int_{-\infty}^{\infty} \Theta(t)e^{i(\omega - \omega_1)t} dt = -\frac{1}{2\pi i} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{e^{-i\omega' t} e^{i(\omega - \omega_1)t}}{\omega' + i\delta} d\omega' dt = \frac{i}{\omega - \omega_1 + i\delta},$$

where we used the following representation for the Heaviside function:

$$\Theta(t) = -\frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{e^{-i\omega t}}{\omega + i\delta} d\omega$$

As we take the limit to the CPA frequency, we see that the response will be given by:

$$\lim_{\omega_1 \rightarrow \omega_{CPA}} \frac{i\rho(\omega_1)}{\omega - \omega_1 + i\delta} \propto \frac{\omega - \omega_{CPA}}{\omega - \omega_{CPA}},$$

as desired.

Derivation of Equation 2

We first consider the effective two reflector system which is composed of the mirror of reflectivity r_1 and the effective reflector composed of the mirror of reflectivity r_2 and the cavity it forms with the perfectly reflecting mirror. We denote the effective reflectivity by $\tilde{r}_2$ and obtain:

$$r = r_1 + t_1 \tilde{r}_2 t_1 e^{2in_1 kl_1} + (t_1 \tilde{r}_2 r_1 \tilde{r}_2 t_1) e^{4in_1 kl_1} = r_1 + t_1^2 \tilde{r}_2 e^{2in_1 kl_1} (1 + e^{2in_1 kl_1} r_1 \tilde{r}_2 + \dots) = r_1 + \frac{t_1^2 \tilde{r}_2 e^{2in_1 kl_1}}{1 - e^{2in_1 kl_1} r_1 \tilde{r}_2},$$

where in the last step we summed a geometric series. To get the final answer, we now need to find $\tilde{r}_2$. Note that in the above, we assumed the reflectivity is the same from both sides of the mirror. This is not generically correct if one side has a different index than the other, so we assume both indices are close to vacuum (it's unclear how much a difference this makes). We now find $\tilde{r}_2$ by noting that the reflectivity of the perfect mirror is -1 (required since $t = 1 + r$ and we want $t = 0$ for a perfect mirror):

$$\tilde{r}_2 = r_2 - \frac{t_2^2 e^{2in_2 kl_2}}{1 + e^{2in_2 kl_2} r_2} = \frac{e^{2in_2 kl_2} (r_2^2 - t_2^2) + r_2}{1 + e^{2in_2 kl_2} r_2} = \frac{r_2 - e^{2in_2 kl_2} (1 + 2r_2)}{1 + e^{2in_2 kl_2} r_2}$$

Note that this is inconsistent with the result in the **Supplement** of the paper (in the section **Calculating a virtual CPA**). Our result agrees only for $2n_2 kl_2 = 2\pi q$ for q an integer. The reason for the discrepancy is that the paper uses incorrect

boundary conditions in this section. The correct boundary conditions are given by (in the notation of that section of the supplement of the paper):

$$\begin{aligned} tB_2 + rA_1 &= A_2, \\ tA_1 + rB_2 &= B_1, \\ B_2e^{-inkl} &= -B_1e^{inkl} \end{aligned}$$

Solving this set of equations gives us the following:

$$tA_1 = B_1 \left[1 + re^{2inkl} \right], A_2 = rA_1 - tB_1e^{2inkl} = A_1 \left[r - \frac{t^2e^{2inkl}}{1 + re^{2inkl}} \right],$$

which is exactly what we derived.

Derivation of Equation 3

As in the paper, we define $x(\omega) = e^{2\pi i k n_2 l_2}$, and $y(\omega) = e^{2\pi i k n_1 l_1}$, where the dependence on frequency is subsumed in the free space wavenumber, $k = \omega/c$. In the below, we define the additional variables, $X(\omega)$ and $Y(\omega)$ as $r(\omega) = X(\omega)/Y(\omega)$. In order to have CPA, we require, first and foremost, the vanishing of the numerator of $r(\omega)$:

$$0 = X(\omega) \equiv r_1(xr_2 + 1) + y(r_2 + x) = 0 \rightarrow y = -r_1 \left[\frac{xr_2 + 1}{r_2 + x} \right],$$

where the form of $X(\omega)$ follows from **Equation 2**. The equation above gives a relation between $x(\omega)$ and $y(\omega)$ at the CPA point. In order to solve for each of these variables individually, we must have another equation. The next equation follows from the requirement $r'(\omega) = 0$. To formulate the relation obtained from this additional requirement, we first calculate the derivatives of $x(\omega)$ and $y(\omega)$:

$$x(\omega) = e^{2in_2\omega l_2/c} \rightarrow x'(\omega) = (2in_2l_2/c)x(\omega), y(\omega) = e^{2in_1\omega l_1/c} \rightarrow y'(\omega) = (2in_1l_1/c)y(\omega)$$

Therefore, we have:

$$X'(\omega) = \frac{2i}{c} \left[r_1 r_2 (n_2 l_2) x + r_2 (n_1 l_1) y + (n_1 l_1 + n_2 l_2) xy \right] =$$

$$\frac{2i n_2 l_2}{c} \left[r_1 r_2 x + r_2 \lambda y + (1 + \lambda) xy \right]$$

Note that requiring $r'(\omega)$ to vanish generally requires

$X'(\omega)Y(\omega) - X(\omega)Y'(\omega) = 0$. But we have already required $X(\omega) = 0$. Thus, we only require $X'(\omega) = 0$ (we obviously cannot have $Y(\omega) = 0$). Thus, we obtain:

$$r_1 r_2 x + r_2 \lambda y + (1 + \lambda) xy = 0 \rightarrow y = -\frac{r_1 r_2 x}{r_2 \lambda + (1 + \lambda)x}$$

We thus have acquired two equations for the two unknowns $x(\omega)$ and $y(\omega)$.

Combining the two equations yields the following:

$$\frac{r_1 r_2 x}{r_2 \lambda + (1 + \lambda)x} = r_1 \left[\frac{x r_2 + 1}{r_2 + x} \right] \rightarrow (r_2 + x) r_1 r_2 x = [r_2 \lambda + (1 + \lambda)x] (r_1 + r_1 r_2 x)$$

Moving all terms to one side and expressing the equation in quadratic form yields:

$$-r_1 r_2 \lambda x^2 + \left[r_1 r_2^2 [1 - \lambda] - r_1 (1 + \lambda) \right] x - r_1 r_2 \lambda = 0$$

Which we may rewrite as (dividing both sides by $-r_1 \lambda$ in order to yield a form that is similar to that of the paper):

$$r_2 x^2 - \left[r_2^2 \left[\frac{1}{\lambda} - 1 \right] - \left[\frac{1}{\lambda} + 1 \right] \right] x + r_2 = 0,$$

which is a quadratic equation with the obvious solution of **Equation 3** of the paper.

Similarly, we may write two equations for $y(\omega)$:

$$x = -\frac{r_2 \lambda y}{r_1 r_2 + (1 + \lambda)y} = -\frac{y r_2 + r_1}{r_1 r_2 + y} \rightarrow r_2 \lambda y [r_1 r_2 + y] = [r_1 r_2 + (1 + \lambda)y] [y r_2 + r_1]$$

We write this in quadratic form as follows:

$$r_2 y^2 - \left[r_1 r_2^2 \left[\lambda - 1 \right] - (1 + \lambda) r_1 \right] y + r_1^2 r_2 = 0$$

From which we obtain:

$$y = r_1 \frac{\left[r_2^2 \left[\lambda - 1 \right] - (1 + \lambda) \right] \pm \sqrt{\left[r_2^2 \left[\lambda - 1 \right] - (1 + \lambda) \right]^2 - 4 r_2^2}}{2 r_2}$$

Note that this differs slightly from **Equation 3** of the paper...